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Final Project Report: Real Estate Agency Portal

Course Project: CIS 344 Database Design & Programming

Semester: Spring 2026

Professor: Yanilda Peralta Ramos

Summary.

The Real Estate Agency Portal is a PHP and MySQL web application designed to simulate a practical real estate agency environment. The system supports three major user roles: agents, buyers, and renters. Agents can manage property listings, while buyers and renters can browse properties, save favorites, and submit inquiries. The project demonstrates important course concepts, including database relationships, primary keys, foreign keys, form handling, sessions, authentication, stored procedures, a database view, a trigger, and a structured web application workflow.

Project Purpose

The purpose of this project is to create a functional real estate web portal that allows different types of users to interact with property records in a realistic way. The portal is not only a frontend website; it is connected to a relational database and includes working back-end logic.

My project focuses on the following major areas:

- PHP front-end supported by HTML and CSS.
- MySQL back-end stores users, properties, inquiries, favorites, and transactions.
- Secure authentication using password hashing, password verification, and sessions.
- Role-based access control for agents, buyers, and renters.
- Image-enabled property listings for a stronger visual detail.
- Database objects such as stored procedures, a view, and a trigger.

Database Design

The database for the project is named `real_estate_portal_db`. It contains the main tables needed for a real estate system: Users, Properties, Inquiries, Transactions, and Favorites. These tables are connected using primary keys and foreign keys so the database can track users, listings, inquiries, saved properties, and completed deals.

The Users table stores account information for agents, buyers, and renters. The Properties table stores listing details and connects each listing to the agent who created it. The Inquiries table stores messages submitted by buyers and renters about specific properties. The Transactions table records completed sale or rental activity. The Favorites table allows buyers and renters to save properties they are interested in.

The database design supports the following relationship logic:

- One agent can create many property listings.
- One buyer or renter can submit many inquiries.
- One property can receive many inquiries from different users.
- One user can save many properties as favorites.
- One property can appear in many users' favorite lists.
- A transaction connects a user and a property when a sale or rental is processed.

Database Objects and SQL Features

My project includes more than basic database tables. It also includes stored procedures, a view, and a trigger. The `AddOrUpdateUser` stored procedure is used to insert a new user or update an existing user. This helps centralize user-related database logic. The `ProcessTransaction` stored procedure is used to record a transaction and update the related property status.

`PropertyListingView` combines property and agent information so listing data can be displayed more easily. The `AfterTransactionInsert` trigger automatically updates the property status after a new transaction is inserted.

Including these SQL features is important because it shows that the project uses database programming concepts, not only simple SELECT and INSERT statements.

```
SELECT
p.propertyId,
p.title,
p.propertyType,
p.address,
p.city,
p.price,
p.status,
u.username AS agentName,
u.contactInfo AS agentContact
FROM Properties p
INNER JOIN Users u
ON p.agentId = u.userId;
```

This JOIN query connects the Properties table with the Users table. The purpose is to display each property listing together with the agent who created it. The relationship works because Properties.agentId is a foreign key that references Users.userId. This query proves that the database uses relationships between tables instead of storing repeated agent information inside the Properties table.

PHP Implementation

The PHP side of the project was completed so the portal works as a functional web application. The database connection uses PDO, which is a stronger and safer approach than directly writing insecure database code. The application logic is organized around the RealEstateDatabase class, where key methods support users, properties, inquiries, favorites, transactions, and filtering.

The login page verifies user credentials and creates a session after successful login.

The register page creates new users and stores passwords with password_hash(). The dashboard page displays user-related activity such as profile information, favorites, inquiries, and

transactions. The Add Property page allows authorized agents to create listings. The properties page displays searchable property cards with images. The property details page shows a larger image and information about one selected property. The submit inquiry page allows buyers or renters to contact the agency about a listing. The favorites page displays saved properties for logged-in users.

Together, these pages demonstrate a complete application workflow from registration and login to browsing, inquiry submission, favorites, and agent listing management.

Authentication and Security

Security is one of the strongest parts of the project. User passwords are not stored as plain text. During registration, the password is processed using `password_hash()`. During login, `password_verify()` checks whether the entered password matches the stored hash.

Property Image Integration

A major improvement in the final project is the addition of property images. The images make the portal look more professional and easier to present. Instead of showing only plain text listing data, the system displays image cards that resemble real real estate platforms.

The images are stored in the `assets/images` directory and connected to sample property records through the `image Path` field in the database. When the SQL file is imported, the sample listings are ready to display images immediately.

Testing and Demonstration Plan

My project can be tested locally using XAMPP. Apache and MySQL should be started from the XAMPP Control Panel. The project folder should be placed inside the `htdocs` directory, and the SQL file should be imported into phpMyAdmin under the `real_estate_portal_db` database.

Demonstration order:

1. Open the homepage and show the featured property image area.
2. Register a new buyer or renter account or log in with a seeded account.
3. Show the secure login process and confirm that the session is created.
4. Log in as an agent and open the dashboard.

5. Open the Add Property page and explain that it is agent-only.
6. Browse the property listings and show the image-enabled cards.
7. Open one property detail page and explain the listing information.
8. Submit an inquiry as a buyer or renter.
9. Save a property to favorites and open the favorites page.
10. Open phpMyAdmin and show the tables, stored procedures, view, and trigger.

Challenges and Solutions

One challenge during development was transforming the starter scaffold into a complete working application. Some original methods and workflows were incomplete, which prevented the portal from behaving like a full system. This was solved by completing the database methods and connecting the PHP pages to the correct SQL logic.

Another challenge was improving the visual quality of the project. A plain text listing page would not look professional during a classroom presentation. This was solved by adding image support, property cards, and sample property images.

A third challenge was improving security. The project needed to avoid unsafe password handling. This was solved by using `password_hash()` during registration and `password_verify()` during login, along with session handling and role-based access checks.

The screen short of my working website

Real Estate Agency Portal

[Home](#) [Properties](#) [Login](#) [Register](#)

WELCOME to IBRAHIM ESTATE PROJECT

This is the completed version of the Real Estate Agency Portal for Spring 2026. **By Professor Yanilda Peralta Ramos**

The platform now supports role-based access

- property browsing, inquiry submission, favorites management
- secure login, with image-rich property listings connected to MySQL



Real Estate Agency Portal

[Home](#) [Properties](#) [Login](#) [Register](#)

Property Listings

Browse the available real estate inventory and open any listing to view details, submit an inquiry, or save it to favorites.

Search by City

Search



Sunset Family Residence

Type: House



Elegant Forest Villa

Type: House



Executive Contemporary Home

Type: House



Family Garden House

Type: House

Real Estate Agency Portal

[Home](#) [Properties](#) [Login](#) [Register](#)

Login

Username

Password

Login

© Spring 2026 Real Estate Agency Portal



Sunset Family Residence

Type: House

Address: 255 Sunset Boulevard

City: Puerto Plata

Price: \$355,000.00

Status: rented

Agent: agent_maria

[View Details](#)



Elegant Forest Villa

Type: House

Address: 17 Pine Crest Lane

City: San Francisco de Macoris

Price: \$398,000.00

Status: available

Agent: agent_maria

[View Details](#)



Executive Contemporary Home

Type: House

Address: 92 Cedar Ridge

City: Punta Cana

Price: \$425,000.00

Status: sold

Agent: agent_maria

[View Details](#)



Family Garden House

Type: House

Address: 154 Green Valley Drive

City: La Vega

Price: \$289,500.00

Status: sold

Agent: agent_maria

[View Details](#)

20

```
21 • CREATE TABLE Users (  
22     userId INT NOT NULL AUTO_INCREMENT,  
23     userName VARCHAR(50) NOT NULL UNIQUE,  
24     contactInfo VARCHAR(200),  
25     passwordHash VARCHAR(255) NOT NULL,  
26     userType ENUM('agent', 'buyer', 'renter') NOT NULL,  
27     PRIMARY KEY (userId)  
28 );  
29  
30 • CREATE TABLE Properties (  
31     propertyId INT NOT NULL AUTO_INCREMENT,  
32     title VARCHAR(100) NOT NULL,  
33     propertyType VARCHAR(50) NOT NULL,  
34     address VARCHAR(200) NOT NULL,  
35     city VARCHAR(100) NOT NULL,  
36     price DECIMAL(12,2) NOT NULL,  
37     status ENUM('available', 'sold', 'rented') NOT NULL DEFAULT 'available',  
38     imagePath VARCHAR(255) NULL,  
39     agentId INT NOT NULL,  
40     PRIMARY KEY (propertyId),  
41     CONSTRAINT fk_properties_agent FOREIGN KEY (agentId) REFERENCES Users(userId)  
42 );  
43  
44 • CREATE TABLE Inquiries (  
45     inquiryId INT NOT NULL AUTO_INCREMENT,  
46     userId INT NOT NULL,  
47     propertyId INT NOT NULL,  
48     message VARCHAR(255) NOT NULL,  
49     inquiryDate DATETIME NOT NULL,  
50     PRIMARY KEY (inquiryId),  
51     CONSTRAINT fk_inquiries_user FOREIGN KEY (userId) REFERENCES Users(userId),  
52     CONSTRAINT fk_inquiries_property FOREIGN KEY (propertyId) REFERENCES Properties(propertyId)  
53 );  
54  
55 • CREATE TABLE Transactions (  
56     transactionId INT NOT NULL AUTO_INCREMENT,  
57     propertyId INT NOT NULL,  
58     userId INT NOT NULL,  
59     transactionType ENUM('sale', 'rental') NOT NULL,  
60     transactionDate DATETIME NOT NULL,  
61     amount DECIMAL(12,2) NOT NULL,  
62     PRIMARY KEY (transactionId),  
63     CONSTRAINT fk_transactions_property FOREIGN KEY (propertyId) REFERENCES Properties(propertyId),  
64     CONSTRAINT fk_transactions_user FOREIGN KEY (userId) REFERENCES Users(userId)  
65 );
```



```

67 CREATE TABLE Favorites (
68     favoriteId INT NOT NULL AUTO_INCREMENT,
69     userId INT NOT NULL,
70     propertyId INT NOT NULL,
71     savedDate DATETIME NOT NULL,
72     PRIMARY KEY (favoriteId),
73     CONSTRAINT fk_favorites_user FOREIGN KEY (userId) REFERENCES Users(userId),
74     CONSTRAINT fk_favorites_property FOREIGN KEY (propertyId) REFERENCES Properties(propertyId),
75     CONSTRAINT uq_favorite_user_property UNIQUE (userId, propertyId)
76 );
77
78 DELIMITER $$
79
80 CREATE PROCEDURE AddOrUpdateUser(
81     IN p_userId INT,
82     IN p_userName VARCHAR(50),
83     IN p_contactInfo VARCHAR(200),
84     IN p_passwordHash VARCHAR(255),
85     IN p_userType ENUM('agent', 'buyer', 'renter')
86 )
87 BEGIN
88     IF p_userId IS NULL OR p_userId = 0 THEN
89         INSERT INTO Users (userName, contactInfo, passwordHash, userType)
90         VALUES (p_userName, p_contactInfo, p_passwordHash, p_userType);

```

Featured Property Experience

The homepage highlights the visual presentation of the portal and introduces visitors to the types of listings available.



Family Homes

Buyers can explore spacious family-oriented houses with detailed pricing, location, and listing information.



Urban Apartments

Renters and buyers can browse modern apartment listings and submit inquiries directly from each property page.



Premium Residences

Agents can present premium listings with image descriptions, and role-based client interaction features.

Real Estate Agency Portal

[Home](#) [Properties](#) [Dashboard](#) [Add Property](#) [Logout](#)

Add Property

Title

Property Type

Address

City

Price

Status

Property Image

No file selected.

Add Property

phpMyAdmin

Showing rows 0 - 6 (7 total, Query took 0.0004 seconds)

SELECT * FROM "properties"

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows Search this table Sort by key: None

Extra options

	propertyId	title	propertyType	address	city	price	status	imagePath	agentId
☐ Edit ☐ Copy ☐ Delete	1	Modern Townhouse Residence	Townhouse	741 Maple Heights	Santo Domingo	215000.00	available	assets/images/house1.jpg	1
☐ Edit ☐ Copy ☐ Delete	2	Luxury Urban Apartment	Apartment	88 Skyline Avenue	Santiago	290000.00	available	assets/images/house4.jpg	1
☐ Edit ☐ Copy ☐ Delete	3	Family Garden House	House	154 Green Valley Drive	Las Vegas	289000.00	sold	assets/images/house2.jpg	1
☐ Edit ☐ Copy ☐ Delete	4	Executive Contemporary Home	House	33 Cedar Ridge	Puerto Rico	435000.00	sold	assets/images/house3.jpg	1
☐ Edit ☐ Copy ☐ Delete	5	Elegant Forest Villa	House	17 Pine Crest Lane	San Francisco de Macoris	330000.00	available	assets/images/house5.jpg	1
☐ Edit ☐ Copy ☐ Delete	6	Stunning Family Residence	House	295 Sunset Boulevard	Puerto Plata	250000.00	rental	assets/images/house6.jpg	1
☐ Edit ☐ Copy ☐ Delete	7	Duplex	House	4034 Baker Street	New York	999000.00	available	N/A	1

phpMyAdmin

Showing rows 0 - 3 (4 total, Query took 0.0002 seconds)

SELECT * FROM "inquiries"

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows Search this table Sort by key: None

Extra options

	InquiryId	userId	propertyId	message	InquiryDate
☐ Edit ☐ Copy ☐ Delete	1	2	1	I would like to schedule a visit this weekend.	2026-03-10 10:00:00
☐ Edit ☐ Copy ☐ Delete	2	3	2	Is this apartment still available for immediate oc...	2026-03-11 14:30:00
☐ Edit ☐ Copy ☐ Delete	3	2	3	Can you share more details about the neighborhood	2026-03-12 09:15:00
☐ Edit ☐ Copy ☐ Delete	4	9	6	Hi, I am interested in this property	2026-05-07 20:56:57

Check all With selected Edit Copy Delete Export

Show all Number of rows: 25 Filter rows Search this table Sort by key: None

Query results operations

phpMyAdmin

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds)

SELECT * FROM "transactions"

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows Search this table Sort by key: None

Extra options

	transactionId	propertyId	userId	transactionType	transactionDate	amount
☐ Edit ☐ Copy ☐ Delete	1	4	2	sale	2026-03-15 11:00:00	425000.00
☐ Edit ☐ Copy ☐ Delete	2	8	3	rental	2026-03-18 16:00:00	355000.00
☐ Edit ☐ Copy ☐ Delete	3	3	2	sale	2026-03-20 13:30:00	289500.00

Check all With selected Edit Copy Delete Export

Show all Number of rows: 25 Filter rows Search this table Sort by key: None